SEC1-SA2 Group 2-Espiritu

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A. Machine Learning Models for Customer Churn

Title: Predicting Customer Churn with Regression-Based and Tree-Based Methods

1. Introduction

Customer churn, also known as customer attrition or turnover, is the percentage of customers who stop doing business, cancel subscriptions, or discontinue using a company's services over a specific period. Predicting customer behavior through data-driven approaches allows companies to identify customers who are likely to cancel their subscriptions and implement retention strategies before churn occurs. This study utilizes the Orange Telecom Churn Dataset, which contains customer usage information and churn labels indicating whether a customer canceled their subscription. The dataset was divided into an 80% training and cross-validation set (churn-bigml-80) and a 20% testing and evaluation set (churn-bigml-20). The primary objective of this study is to build and evaluate regression-based and tree-based machine learning models capable of predicting customer churn and comparing their predictive performance.

2. Methodology

2.1 Data Description

```r
# Load training and testing datasets
churn_train <- read.csv("churn-bigml-80.csv")
churn_test  <- read.csv("churn-bigml-20.csv")

# Dimensions
dim(churn_train)
```

```
## [1] 2666   20
```

```r
dim(churn_test)
```

```
## [1] 667  20
```

```r
# Display Variable Types
str(churn_train)
```

```
## 'data.frame':    2666 obs. of  20 variables:
##  $ State                 : chr  "KS" "OH" "NJ" "OH" ...
##  $ Account.length        : int  128 107 137 84 75 118 121 147 141 74 ...
##  $ Area.code             : int  415 415 415 408 415 510 510 415 415 415 ...
##  $ International.plan    : chr  "No" "No" "No" "Yes" ...
##  $ Voice.mail.plan       : chr  "Yes" "Yes" "No" "No" ...
##  $ Number.vmail.messages : int  25 26 0 0 0 0 24 0 37 0 ...
##  $ Total.day.minutes     : num  265 162 243 299 167 ...
```

```
##  $ Total.day.calls       : int  110 123 114 71 113 98 88 79 84 127 ...
##  $ Total.day.charge      : num  45.1 27.5 41.4 50.9 28.3 ...
##  $ Total.eve.minutes     : num  197.4 195.5 121.2 61.9 148.3 ...
##  $ Total.eve.calls       : int  99 103 110 88 122 101 108 94 111 148 ...
##  $ Total.eve.charge      : num  16.78 16.62 10.3 5.26 12.61 ...
##  $ Total.night.minutes   : num  245 254 163 197 187 ...
##  $ Total.night.calls     : int  91 103 104 89 121 118 118 96 97 94 ...
##  $ Total.night.charge    : num  11.01 11.45 7.32 8.86 8.41 ...
##  $ Total.intl.minutes    : num  10 13.7 12.2 6.6 10.1 6.3 7.5 7.1 11.2 9.1 ...
##  $ Total.intl.calls      : int  3 3 5 7 3 6 7 6 5 5 ...
##  $ Total.intl.charge     : num  2.7 3.7 3.29 1.78 2.73 1.7 2.03 1.92 3.02 2.46 ...
##  $ Customer.service.calls: int  1 1 0 2 3 0 3 0 0 0 ...
##  $ Churn                 : chr  "False" "False" "False" "False" ...
```

```r
str(churn_test)
```

```
## 'data.frame':    667 obs. of  20 variables:
##  $ State                 : chr  "LA" "IN" "NY" "SC" ...
##  $ Account.length        : int  117 65 161 111 49 36 65 119 10 68 ...
##  $ Area.code             : int  408 415 415 415 510 408 415 415 408 415 ...
##  $ International.plan    : chr  "No" "No" "No" "No" ...
##  $ Voice.mail.plan       : chr  "No" "No" "No" "No" ...
##  $ Number.vmail.messages : int  0 0 0 0 0 30 0 0 0 0 ...
##  $ Total.day.minutes     : num  184 129 333 110 119 ...
##  $ Total.day.calls       : int  97 137 67 103 117 128 120 114 112 70 ...
##  $ Total.day.charge      : num  31.4 21.9 56.6 18.8 20.3 ...
##  $ Total.eve.minutes     : num  352 228 318 137 215 ...
##  $ Total.eve.calls       : int  80 83 97 102 109 80 122 117 66 164 ...
##  $ Total.eve.charge      : num  29.9 19.4 27 11.7 18.3 ...
##  $ Total.night.minutes   : num  216 209 161 190 179 ...
##  $ Total.night.calls     : int  90 111 128 105 90 109 118 91 57 103 ...
##  $ Total.night.charge    : num  9.71 9.4 7.23 8.53 8.04 ...
##  $ Total.intl.minutes    : num  8.7 12.7 5.4 7.7 11.1 14.5 13.2 8.8 11.4 12.1 ...
##  $ Total.intl.calls      : int  4 6 9 6 1 6 5 3 6 3 ...
##  $ Total.intl.charge     : num  2.35 3.43 1.46 2.08 3 3.92 3.56 2.38 3.08 3.27 ...
##  $ Customer.service.calls: int  1 4 4 2 1 0 3 5 2 3 ...
##  $ Churn                 : chr  "False" "True" "True" "False" ...
```

The training dataset (churn_train) contains 2,666 observations and 20 variables, while the testing dataset (churn_test) contains 667 observations and 20 variables, reflecting the intended 80/20 dataset split for machine learning model development and evaluation. The dataset consists of both numerical and categorical variables related to customer account information, call usage behavior, service plans, and customer service interactions, with the Churn variable serving as the target classification outcome.

```r
# Summary Statistics
summary(churn_train)
```

```
##     State           Account.length    Area.code     International.plan
##  Length:2666        Min.   :  1.0   Min.   :408.0   Length:2666       
##  Class :character   1st Qu.: 73.0   1st Qu.:408.0   Class :character  
##  Mode  :character   Median :100.0   Median :415.0   Mode  :character  
##                     Mean   :100.6   Mean   :437.4                     
##                     3rd Qu.:127.0   3rd Qu.:510.0                     
##                     Max.   :243.0   Max.   :510.0                     
##  Voice.mail.plan    Number.vmail.messages Total.day.minutes Total.day.calls
##  Length:2666        Min.   : 0.000        Min.   :  0.0     Min.   :  0.0  
```

```
##  Class :character   1st Qu.: 0.000        1st Qu.:143.4     1st Qu.: 87.0
##  Mode  :character   Median : 0.000        Median :179.9     Median :101.0
##                     Mean   : 8.022        Mean   :179.5     Mean   :100.3
##                     3rd Qu.:19.000        3rd Qu.:215.9     3rd Qu.:114.0
##                     Max.   :50.000        Max.   :350.8     Max.   :160.0
##  Total.day.charge Total.eve.minutes Total.eve.calls Total.eve.charge
##  Min.   : 0.00    Min.   :  0.0     Min.   :  0     Min.   : 0.00
##  1st Qu.:24.38    1st Qu.:165.3     1st Qu.: 87     1st Qu.:14.05
##  Median :30.59    Median :200.9     Median :100     Median :17.08
##  Mean   :30.51    Mean   :200.4     Mean   :100     Mean   :17.03
##  3rd Qu.:36.70    3rd Qu.:235.1     3rd Qu.:114     3rd Qu.:19.98
##  Max.   :59.64    Max.   :363.7     Max.   :170     Max.   :30.91
##  Total.night.minutes Total.night.calls Total.night.charge Total.intl.minutes
##  Min.   : 43.7       Min.   : 33.0     Min.   : 1.970     Min.   : 0.00
##  1st Qu.:166.9       1st Qu.: 87.0     1st Qu.: 7.513     1st Qu.: 8.50
##  Median :201.2       Median :100.0     Median : 9.050     Median :10.20
##  Mean   :201.2       Mean   :100.1     Mean   : 9.053     Mean   :10.24
##  3rd Qu.:236.5       3rd Qu.:113.0     3rd Qu.:10.640     3rd Qu.:12.10
##  Max.   :395.0       Max.   :166.0     Max.   :17.770     Max.   :20.00
##  Total.intl.calls Total.intl.charge Customer.service.calls    Churn
##  Min.   : 0.000   Min.   :0.000     Min.   :0.000          Length:2666
##  1st Qu.: 3.000   1st Qu.:2.300     1st Qu.:1.000          Class :character
##  Median : 4.000   Median :2.750     Median :1.000          Mode  :character
##  Mean   : 4.467   Mean   :2.764     Mean   :1.563
##  3rd Qu.: 6.000   3rd Qu.:3.270     3rd Qu.:2.000
##  Max.   :20.000   Max.   :5.400     Max.   :9.000
```

```r
summary(churn_test)
```

```
##     State           Account.length    Area.code     International.plan
##  Length:667         Min.   :  1.0   Min.   :408.0   Length:667
##  Class :character   1st Qu.: 76.0   1st Qu.:408.0   Class :character
##  Mode  :character   Median :102.0   Median :415.0   Mode  :character
##                     Mean   :102.8   Mean   :436.2
##                     3rd Qu.:128.0   3rd Qu.:415.0
##                     Max.   :232.0   Max.   :510.0
##  Voice.mail.plan    Number.vmail.messages Total.day.minutes Total.day.calls
##  Length:667         Min.   : 0.000        Min.   : 25.9     Min.   : 30.0
##  Class :character   1st Qu.: 0.000        1st Qu.:146.2     1st Qu.: 87.5
##  Mode  :character   Median : 0.000        Median :178.3     Median :101.0
##                     Mean   : 8.408        Mean   :180.9     Mean   :100.9
##                     3rd Qu.:20.000        3rd Qu.:220.7     3rd Qu.:115.0
##                     Max.   :51.000        Max.   :334.3     Max.   :165.0
##  Total.day.charge Total.eve.minutes Total.eve.calls Total.eve.charge
##  Min.   : 4.40    Min.   : 48.1     Min.   : 37.0   Min.   : 4.09
##  1st Qu.:24.86    1st Qu.:171.1     1st Qu.: 88.0   1st Qu.:14.54
##  Median :30.31    Median :203.7     Median :101.0   Median :17.31
##  Mean   :30.76    Mean   :203.4     Mean   :100.5   Mean   :17.29
##  3rd Qu.:37.52    3rd Qu.:236.4     3rd Qu.:113.0   3rd Qu.:20.09
##  Max.   :56.83    Max.   :361.8     Max.   :168.0   Max.   :30.75
##  Total.night.minutes Total.night.calls Total.night.charge Total.intl.minutes
##  Min.   : 23.2       Min.   : 42.0     Min.   : 1.040     Min.   : 0.00
##  1st Qu.:167.9       1st Qu.: 86.0     1st Qu.: 7.560     1st Qu.: 8.60
##  Median :201.6       Median :100.0     Median : 9.070     Median :10.50
##  Mean   :199.7       Mean   :100.1     Mean   : 8.986     Mean   :10.24
```

```
##  3rd Qu.:231.5       3rd Qu.:113.5     3rd Qu.:10.420     3rd Qu.:12.05
##  Max.   :367.7       Max.   :175.0     Max.   :16.550     Max.   :18.30
##  Total.intl.calls Total.intl.charge Customer.service.calls    Churn
##  Min.   : 0.000   Min.   :0.000     Min.   :0.000          Length:667
##  1st Qu.: 3.000   1st Qu.:2.320     1st Qu.:1.000          Class :character
##  Median : 4.000   Median :2.840     Median :1.000          Mode  :character
##  Mean   : 4.528   Mean   :2.765     Mean   :1.564
##  3rd Qu.: 6.000   3rd Qu.:3.255     3rd Qu.:2.000
##  Max.   :18.000   Max.   :4.940     Max.   :8.000
```

The summary statistics indicate that the training and testing datasets have relatively similar distributions across numerical variables, suggesting that the 80/20 split was balanced and appropriate for machine learning analysis. Variables such as total day minutes, evening minutes, night minutes, and customer service calls show noticeable variability among customers, which may contribute significantly to predicting customer churn behavior.

2.2 Data Preprocessing

```r
## Remove Non-Predictive Identifier Variables
# Remove identifiers
churn_train <- churn_train %>%
  dplyr::select(-State, -Area.code)

churn_test <- churn_test %>%
  dplyr::select(-State, -Area.code)

# Verify removal
colnames(churn_train)
```

```
##  [1] "Account.length"         "International.plan"     "Voice.mail.plan"
##  [4] "Number.vmail.messages"  "Total.day.minutes"      "Total.day.calls"
##  [7] "Total.day.charge"       "Total.eve.minutes"      "Total.eve.calls"
## [10] "Total.eve.charge"       "Total.night.minutes"    "Total.night.calls"
## [13] "Total.night.charge"     "Total.intl.minutes"     "Total.intl.calls"
## [16] "Total.intl.charge"      "Customer.service.calls" "Churn"
```

```r
colnames(churn_test)
```

```
##  [1] "Account.length"         "International.plan"     "Voice.mail.plan"
##  [4] "Number.vmail.messages"  "Total.day.minutes"      "Total.day.calls"
##  [7] "Total.day.charge"       "Total.eve.minutes"      "Total.eve.calls"
## [10] "Total.eve.charge"       "Total.night.minutes"    "Total.night.calls"
## [13] "Total.night.charge"     "Total.intl.minutes"     "Total.intl.calls"
## [16] "Total.intl.charge"      "Customer.service.calls" "Churn"
```

```r
## Convert Churn into Binary Variable

# Checking
table(churn_train$Churn)
```

```
##
## False  True
##  2278   388
```

```r
table(churn_test$Churn)
```

```
##
```

```
## False  True
##   572    95
```

```r
#Convert into binary factor values.
churn_train$Churn <- factor(
  churn_train$Churn,
  levels = c("False", "True"),
  labels = c(0, 1)
)

churn_test$Churn <- factor(
  churn_test$Churn,
  levels = c("False", "True"),
  labels = c(0, 1)
)

# Verify factor conversion
table(churn_train$Churn)
```

```
##
##    0    1
## 2278  388
```

```r
table(churn_test$Churn)
```

```
##
##   0   1
## 572  95
```

```r
str(churn_train$Churn)
```

```
##  Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
```

```r
str(churn_test$Churn)
```

```
##  Factor w/ 2 levels "0","1": 1 2 2 1 1 1 1 2 1 1 ...
```

```r
# Convert categorical variables into factors
churn_train$International.plan <- as.factor(churn_train$International.plan)
churn_train$Voice.mail.plan <- as.factor(churn_train$Voice.mail.plan)

churn_test$International.plan <- as.factor(churn_test$International.plan)
churn_test$Voice.mail.plan <- as.factor(churn_test$Voice.mail.plan)

# Verify factor conversion
str(churn_train$International.plan)
```

```
##  Factor w/ 2 levels "No","Yes": 1 1 1 2 2 2 1 2 2 1 ...
```

```r
str(churn_train$Voice.mail.plan)
```

```
##  Factor w/ 2 levels "No","Yes": 2 2 1 1 1 1 2 1 2 1 ...
```

```r
str(churn_test$International.plan)
```

```
##  Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1 ...
```

```r
str(churn_test$Voice.mail.plan)
```

```
##  Factor w/ 2 levels "No","Yes": 1 1 1 1 1 2 1 1 1 1 ...
```

```r
# Apply One-Hot Encoding

#This converts factor variables into machine-learning-friendly numeric columns.
# Create dummy variable model
dummy_model <- dummyVars(Churn ~ ., data = churn_train)

# Apply encoding
train_encoded <- stats::predict(dummy_model, newdata = churn_train)
test_encoded  <- stats::predict(dummy_model, newdata = churn_test)

# Convert to data frames
train_encoded <- as.data.frame(train_encoded)
test_encoded  <- as.data.frame(test_encoded)

# Add target variable back
train_encoded$Churn <- churn_train$Churn
test_encoded$Churn  <- churn_test$Churn

# Preview encoded data
head(train_encoded)
```

```
##   Account.length International.plan.No International.plan.Yes
## 1            128                     1                      0
## 2            107                     1                      0
## 3            137                     1                      0
## 4             84                     0                      1
## 5             75                     0                      1
## 6            118                     0                      1
##   Voice.mail.plan.No Voice.mail.plan.Yes Number.vmail.messages
## 1                  0                   1                    25
## 2                  0                   1                    26
## 3                  1                   0                     0
## 4                  1                   0                     0
## 5                  1                   0                     0
## 6                  1                   0                     0
##   Total.day.minutes Total.day.calls Total.day.charge Total.eve.minutes
## 1             265.1             110            45.07             197.4
## 2             161.6             123            27.47             195.5
## 3             243.4             114            41.38             121.2
## 4             299.4              71            50.90              61.9
## 5             166.7             113            28.34             148.3
## 6             223.4              98            37.98             220.6
##   Total.eve.calls Total.eve.charge Total.night.minutes Total.night.calls
## 1              99            16.78               244.7                91
## 2             103            16.62               254.4               103
## 3             110            10.30               162.6               104
## 4              88             5.26               196.9                89
## 5             122            12.61               186.9               121
## 6             101            18.75               203.9               118
##   Total.night.charge Total.intl.minutes Total.intl.calls Total.intl.charge
## 1              11.01               10.0                3              2.70
## 2              11.45               13.7                3              3.70
## 3               7.32               12.2                5              3.29
## 4               8.86                6.6                7              1.78
```

```
## 5             8.41            10.1                3               2.73
## 6             9.18             6.3                6               1.70
##   Customer.service.calls Churn
## 1                      1     0
## 2                      1     0
## 3                      0     0
## 4                      2     0
## 5                      3     0
## 6                      0     0
```

```r
head(test_encoded)
```

```
##   Account.length International.plan.No International.plan.Yes
## 1            117                    1                      0
## 2             65                    1                      0
## 3            161                    1                      0
## 4            111                    1                      0
## 5             49                    1                      0
## 6             36                    1                      0
##   Voice.mail.plan.No Voice.mail.plan.Yes Number.vmail.messages
## 1                  1                   0                     0
## 2                  1                   0                     0
## 3                  1                   0                     0
## 4                  1                   0                     0
## 5                  1                   0                     0
## 6                  0                   1                    30
##   Total.day.minutes Total.day.calls Total.day.charge Total.eve.minutes
## 1             184.5              97            31.37             351.6
## 2             129.1             137            21.95             228.5
## 3             332.9              67            56.59             317.8
## 4             110.4             103            18.77             137.3
## 5             119.3             117            20.28             215.1
## 6             146.3             128            24.87             162.5
##   Total.eve.calls Total.eve.charge Total.night.minutes Total.night.calls
## 1              80            29.89               215.8                90
## 2              83            19.42               208.8               111
## 3              97            27.01               160.6               128
## 4             102            11.67               189.6               105
## 5             109            18.28               178.7                90
## 6              80            13.81               129.3               109
##   Total.night.charge Total.intl.minutes Total.intl.calls Total.intl.charge
## 1               9.71                8.7                4              2.35
## 2               9.40               12.7                6              3.43
## 3               7.23                5.4                9              1.46
## 4               8.53                7.7                6              2.08
## 5               8.04               11.1                1              3.00
## 6               5.82               14.5                6              3.92
##   Customer.service.calls Churn
## 1                      1     0
## 2                      4     1
## 3                      4     1
## 4                      2     0
## 5                      1     0
## 6                      0     0
```

```r
# Structure of encoded training data
str(train_encoded)
```

```
## 'data.frame':    2666 obs. of  20 variables:
##  $ Account.length        : num  128 107 137 84 75 118 121 147 141 74 ...
##  $ International.plan.No : num  1 1 1 0 0 0 1 0 0 1 ...
##  $ International.plan.Yes: num  0 0 0 1 1 1 0 1 1 0 ...
##  $ Voice.mail.plan.No    : num  0 0 1 1 1 1 0 1 0 1 ...
##  $ Voice.mail.plan.Yes   : num  1 1 0 0 0 0 1 0 1 0 ...
##  $ Number.vmail.messages : num  25 26 0 0 0 0 24 0 37 0 ...
##  $ Total.day.minutes     : num  265 162 243 299 167 ...
##  $ Total.day.calls       : num  110 123 114 71 113 98 88 79 84 127 ...
##  $ Total.day.charge      : num  45.1 27.5 41.4 50.9 28.3 ...
##  $ Total.eve.minutes     : num  197.4 195.5 121.2 61.9 148.3 ...
##  $ Total.eve.calls       : num  99 103 110 88 122 101 108 94 111 148 ...
##  $ Total.eve.charge      : num  16.78 16.62 10.3 5.26 12.61 ...
##  $ Total.night.minutes   : num  245 254 163 197 187 ...
##  $ Total.night.calls     : num  91 103 104 89 121 118 118 96 97 94 ...
##  $ Total.night.charge    : num  11.01 11.45 7.32 8.86 8.41 ...
##  $ Total.intl.minutes    : num  10 13.7 12.2 6.6 10.1 6.3 7.5 7.1 11.2 9.1 ...
##  $ Total.intl.calls      : num  3 3 5 7 3 6 7 6 5 5 ...
##  $ Total.intl.charge     : num  2.7 3.7 3.29 1.78 2.73 1.7 2.03 1.92 3.02 2.46 ...
##  $ Customer.service.calls: num  1 1 0 2 3 0 3 0 0 0 ...
##  $ Churn                 : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
```

```r
str(test_encoded)
```

```
## 'data.frame':    667 obs. of  20 variables:
##  $ Account.length        : num  117 65 161 111 49 36 65 119 10 68 ...
##  $ International.plan.No : num  1 1 1 1 1 1 1 1 1 1 ...
##  $ International.plan.Yes: num  0 0 0 0 0 0 0 0 0 0 ...
##  $ Voice.mail.plan.No    : num  1 1 1 1 1 0 1 1 1 1 ...
##  $ Voice.mail.plan.Yes   : num  0 0 0 0 0 1 0 0 0 0 ...
##  $ Number.vmail.messages : num  0 0 0 0 0 30 0 0 0 0 ...
##  $ Total.day.minutes     : num  184 129 333 110 119 ...
##  $ Total.day.calls       : num  97 137 67 103 117 128 120 114 112 70 ...
##  $ Total.day.charge      : num  31.4 21.9 56.6 18.8 20.3 ...
##  $ Total.eve.minutes     : num  352 228 318 137 215 ...
##  $ Total.eve.calls       : num  80 83 97 102 109 80 122 117 66 164 ...
##  $ Total.eve.charge      : num  29.9 19.4 27 11.7 18.3 ...
##  $ Total.night.minutes   : num  216 209 161 190 179 ...
##  $ Total.night.calls     : num  90 111 128 105 90 109 118 91 57 103 ...
##  $ Total.night.charge    : num  9.71 9.4 7.23 8.53 8.04 ...
##  $ Total.intl.minutes    : num  8.7 12.7 5.4 7.7 11.1 14.5 13.2 8.8 11.4 12.1 ...
##  $ Total.intl.calls      : num  4 6 9 6 1 6 5 3 6 3 ...
##  $ Total.intl.charge     : num  2.35 3.43 1.46 2.08 3 3.92 3.56 2.38 3.08 3.27 ...
##  $ Customer.service.calls: num  1 4 4 2 1 0 3 5 2 3 ...
##  $ Churn                 : Factor w/ 2 levels "0","1": 1 2 2 1 1 1 1 2 1 1 ...
```

```r
# Dataset dimensions
dim(train_encoded)
```

```
## [1] 2666   20
```

```r
dim(test_encoded)
```

```
## [1] 667  20
```

```r
# Check missing values
sum(is.na(train_encoded))
```

```
## [1] 0
```

```r
sum(is.na(test_encoded))
```

```
## [1] 0
```

The dataset underwent several preprocessing procedures to prepare the data for machine learning analysis. Non-predictive identifier variables such as State and Area.code were removed to reduce irrelevant information, while the target variable Churn was converted into a binary factor variable where 0 represented non-churn customers and 1 represented churn customers. Additionally, categorical variables were transformed into factor data types and encoded using one-hot encoding to convert them into numerical representations compatible with machine learning algorithms, followed by missing value verification to ensure data quality and completeness.

2.3 Exploratory Data Analysis

```r
# A. Churn Rate Analysis

## Frequency Table
table(churn_train$Churn)
```

```
##
##    0    1
## 2278  388
```

```r
prop.table(table(churn_train$Churn))
```

```
##
##         0         1
## 0.8544636 0.1455364
```

```r
# Bar Plot of Churn Distribution
ggplot(churn_train, aes(x = Churn)) +
  geom_bar(fill = "steelblue") +

  geom_text(
    stat = "count",
    aes(label = after_stat(count)),
    vjust = -0.5
  ) +
  scale_x_discrete(
    labels = c(
      "0" = "Retained Customer",
      "1" = "Customer Left"
    )
  ) +
  ylim(0, 2500) +
  labs(
    title = "Figure 1. Distribution of Customer Retention Status",
    x = "Customer Status",
    y = "Frequency"
  ) +
```

```
theme_minimal()
```

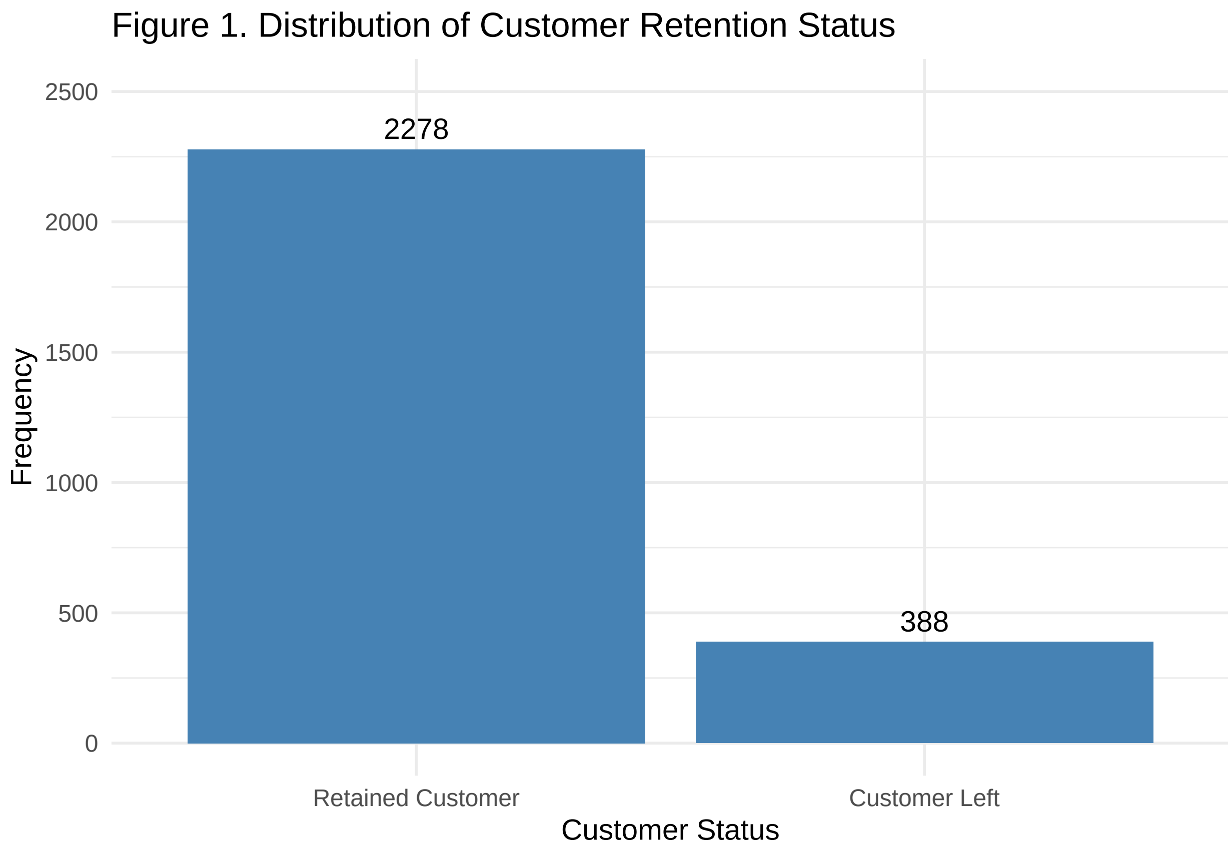


Figure 1 shows that the majority of customers were retained, with 2,278 observations (85.45%), while only 388 customers (14.55%) left the subscription service. This indicates the presence of class imbalance within the dataset, which may influence machine learning model performance by favoring predictions toward the majority retained-customer class.

```
numeric_data <- train_encoded %>%
  dplyr::select(where(is.numeric))

# Correlation matrix
cor_matrix <- cor(numeric_data)

# Correlation plot
corrplot(
  cor_matrix,
  method = "color",
  type = "upper",

  col = colorRampPalette(c("red", "white", "blue"))(200),

  addCoef.col = "black",

  tl.col = "black",
  tl.cex = 0.7,
  number.cex = 0.5,
```

```r
  tl.srt = 45,

  order = "hclust",
  diag = FALSE,

  main = "Figure 2. Correlation Heatmap of Numerical Variables",

  mar = c(0, 0, 2, 0)
)
```

Figure 2. Correlation Heatmap of Numerical Variables

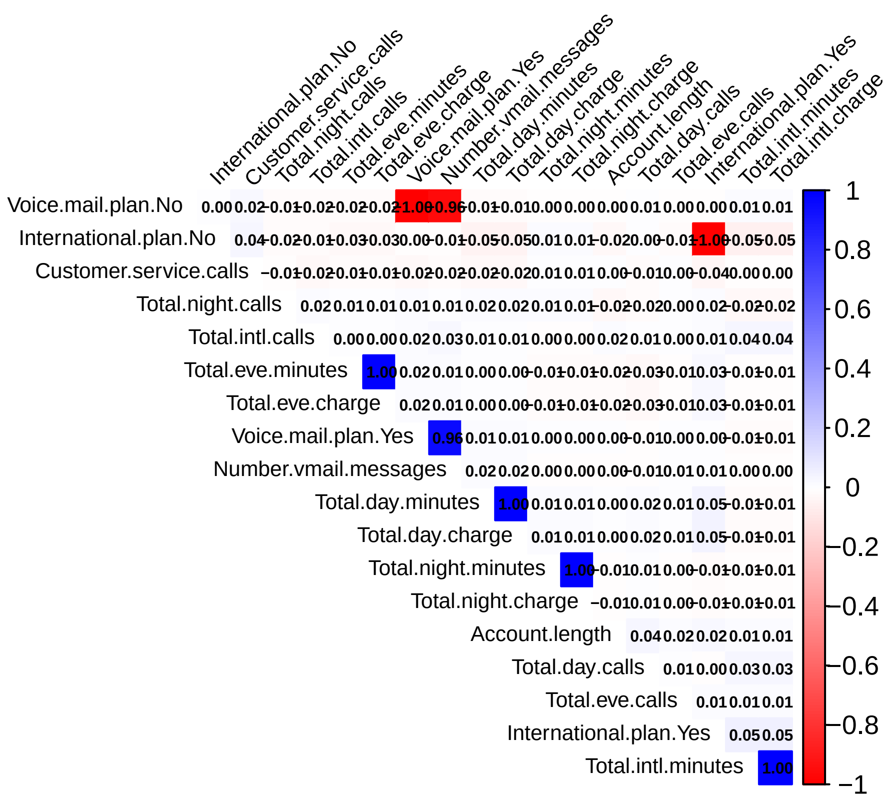


Figure 2 shows that several variables have extremely high positive correlations, particularly between call minute variables and their corresponding charge variables such as Total.day.minutes and Total.day.charge, because charges are directly calculated from usage minutes. Additionally, the strong correlation between encoded categorical pairs such as Voice.mail.plan.Yes and Voice.mail.plan.No is expected due to one-hot encoding, where the presence of one category automatically implies the absence of the other category.

```r
# Boxplot of Customer Service Calls by Churn Status
ggplot(churn_train,
       aes(x = Churn,
           y = Customer.service.calls,
           fill = Churn)) +

  geom_boxplot() +

  scale_x_discrete(
    labels = c(
      "0" = "Retained Customer",
```

```r
      "1" = "Customer Left"
    )
  ) +

  labs(
    title = "Figure 3. Customer Service Calls by Customer Status",
    x = "Customer Status",
    y = "Number of Customer Service Calls"
  ) +

  theme_minimal() +
  theme(
    legend.position = "none"
  )
```

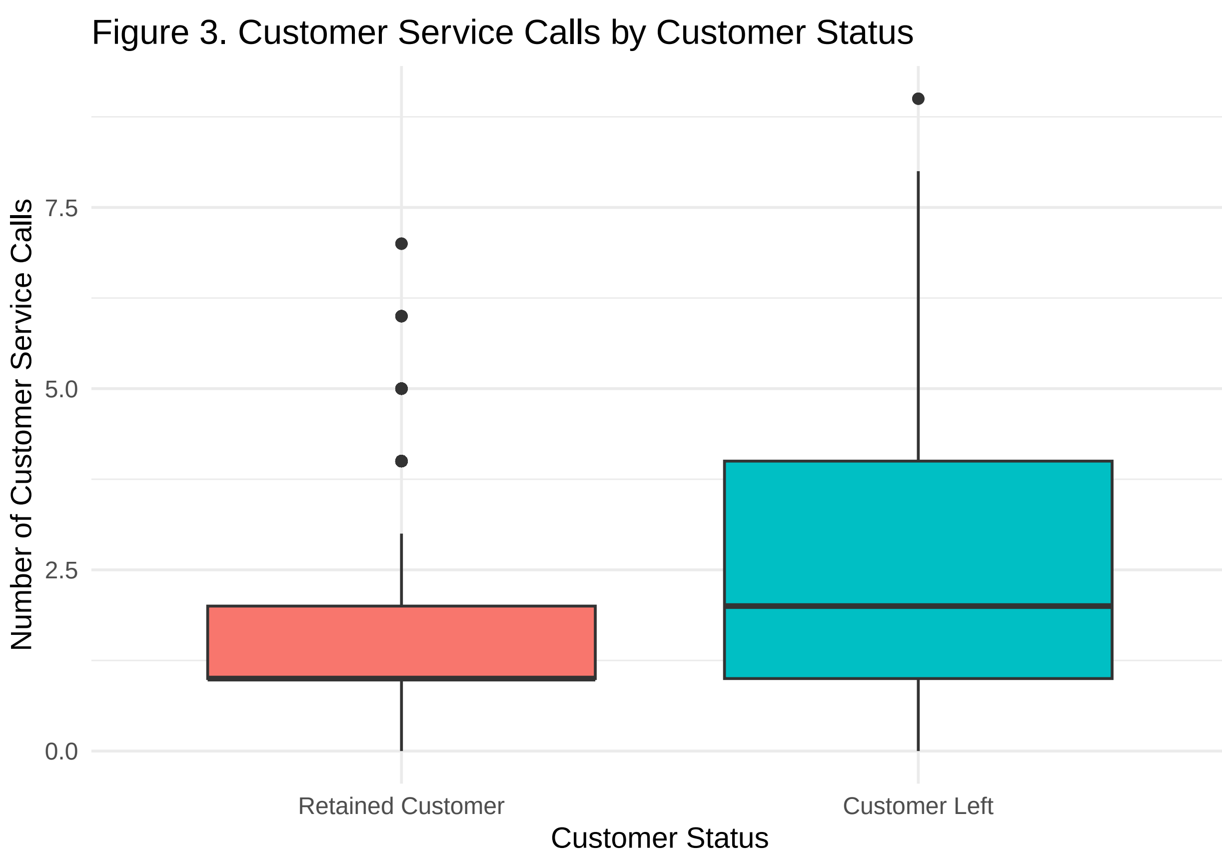


Figure 3 indicates that customers who left the subscription service generally made more customer service calls compared to retained customers. The higher median and wider spread among customers who left suggest that frequent interactions with customer support may be associated with customer dissatisfaction and an increased likelihood of churn.

```r
# Distribution of Total Day Minutes
ggplot(churn_train,
       aes(x = Total.day.minutes)) +

  geom_histogram(
    bins = 20,
    fill = "steelblue",
```

```r
    color = "black"
  ) +

  labs(
    title = "Figure 4. Distribution of Total Day Minutes",
    x = "Total Day Minutes",
    y = "Frequency"
  ) +

  theme_minimal()
```

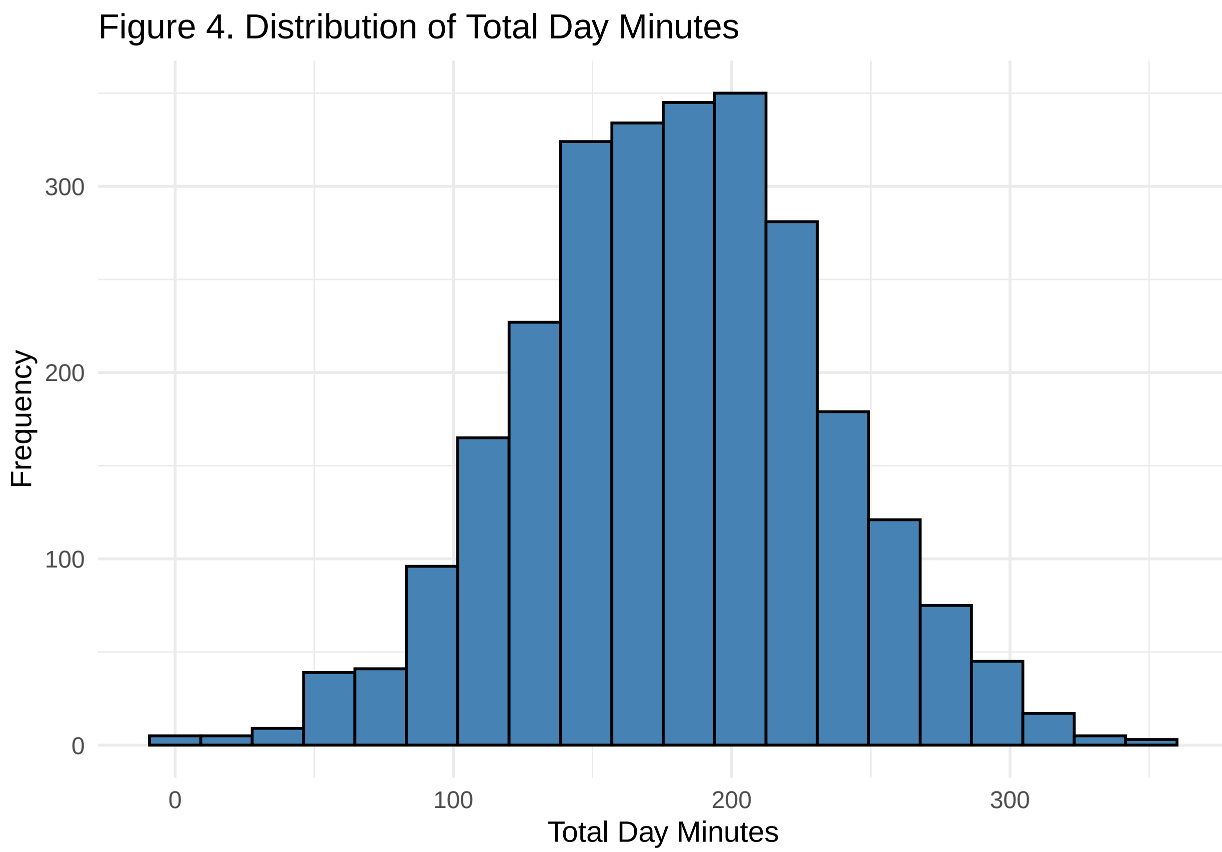


Figure 4 represents the total number of daytime call minutes used by a customer during the billing period. Its shows that the distribution of total day minutes is approximately bell-shaped, with most customers using between 100 and 250 total daytime minutes. The presence of a few customers with extremely low or high usage values suggests moderate variability in customer calling behavior, which may influence churn prediction patterns.

```r
# International Plan vs Churn
ggplot(churn_train,
       aes(x = International.plan,
           fill = Churn)) +

  geom_bar(position = "fill") +

  scale_fill_discrete(
    labels = c(
      "Retained Customer",
```

```
      "Customer Left"
    )
  ) +

  labs(
    title = "Figure 5. Customer Status by International Plan",
    x = "International Plan",
    y = "Proportion",
    fill = "Customer Status"
  ) +

  theme_minimal()
```

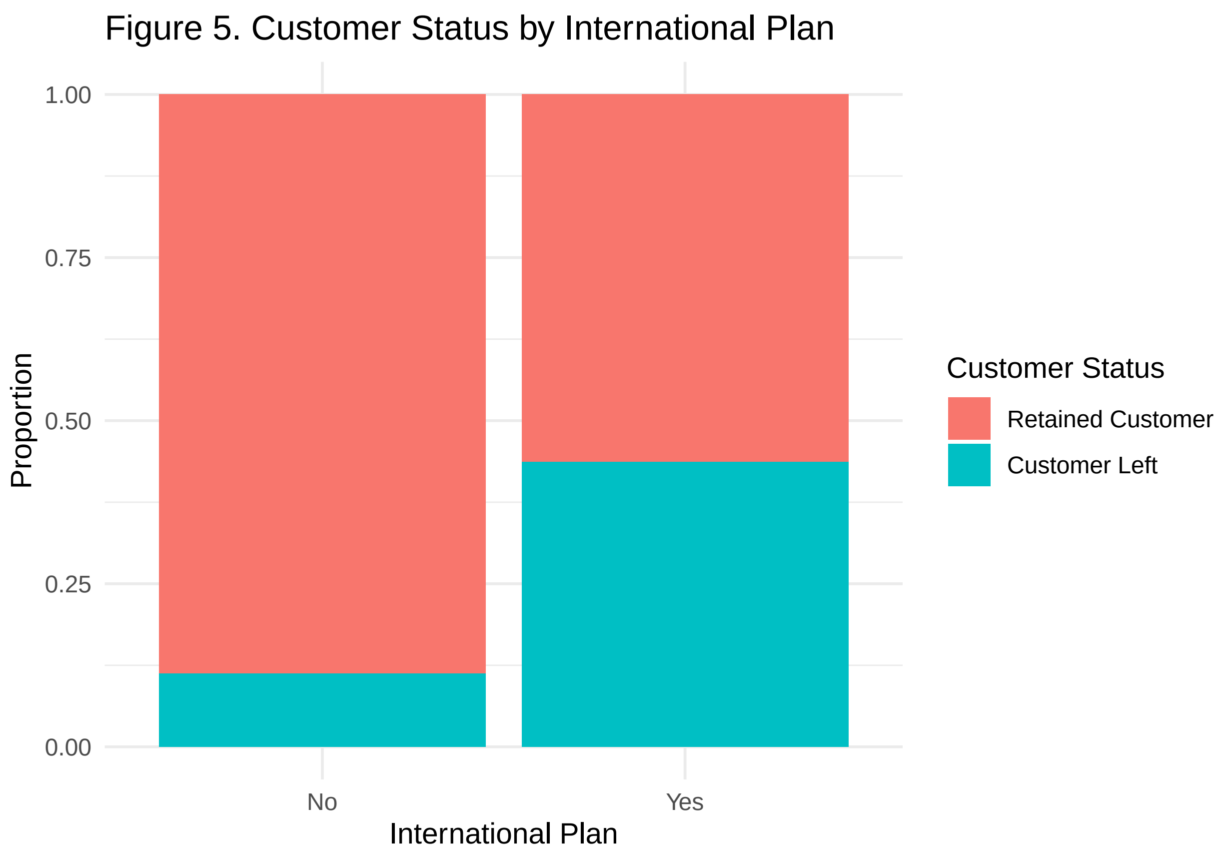


Figure 5 indicates that customers subscribed to an international plan exhibited a noticeably higher proportion of customer departures compared to customers without an international plan. This suggests that having an international plan may be associated with an increased likelihood of churn and could serve as an important predictor variable in the machine learning models.

2.4 Model Building

Logistic Regression was selected as a baseline regression-based classification model because it is widely used for binary outcome prediction problems such as customer churn classification. The model estimates the probability of customer churn based on predictor variables and provides interpretable coefficients that help identify factors associated with customer retention and cancellation behavior.

```
# Logistic Regression Model
log_model <- glm(
```

```r
  Churn ~ .,
  data = train_encoded,
  family = binomial
)

# Model summary
summary(log_model)
```

```
##
## Call:
## glm(formula = Churn ~ ., family = binomial, data = train_encoded)
##
## Coefficients: (2 not defined because of singularities)
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)            -7.973e+00  1.055e+00  -7.557 4.13e-14 ***
## Account.length          9.048e-04  1.567e-03   0.577  0.56366
## International.plan.No  -2.101e+00  1.597e-01 -13.159  < 2e-16 ***
## International.plan.Yes         NA         NA      NA       NA
## Voice.mail.plan.No      2.068e+00  6.567e-01   3.149  0.00164 **
## Voice.mail.plan.Yes            NA         NA      NA       NA
## Number.vmail.messages   3.860e-02  2.065e-02   1.869  0.06162 .
## Total.day.minutes      -5.051e-01  3.671e+00  -0.138  0.89057
## Total.day.calls         2.874e-03  3.104e-03   0.926  0.35438
## Total.day.charge        3.045e+00  2.159e+01   0.141  0.88786
## Total.eve.minutes       1.563e+00  1.842e+00   0.849  0.39612
## Total.eve.calls        -7.707e-04  3.085e-03  -0.250  0.80273
## Total.eve.charge       -1.832e+01  2.167e+01  -0.845  0.39784
## Total.night.minutes    -7.361e-03  9.821e-01  -0.007  0.99402
## Total.night.calls       1.990e-03  3.170e-03   0.628  0.53009
## Total.night.charge      2.257e-01  2.182e+01   0.010  0.99175
## Total.intl.minutes     -3.617e+00  5.935e+00  -0.609  0.54223
## Total.intl.calls       -1.206e-01  2.887e-02  -4.176 2.97e-05 ***
## Total.intl.charge       1.377e+01  2.198e+01   0.626  0.53107
## Customer.service.calls  5.072e-01  4.410e-02  11.500  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2212.2  on 2665  degrees of freedom
## Residual deviance: 1729.6  on 2648  degrees of freedom
## AIC: 1765.6
##
## Number of Fisher Scoring iterations: 6
```

The logistic regression results indicate that several predictor variables significantly influenced customer churn behavior. Variables such as International.plan.No, Voice.mail.plan.No, Total.intl.calls, and Customer.service.calls were statistically significant predictors, suggesting that customer subscription features, international calling behavior, and frequent customer service interactions are associated with churn probability. In particular, the positive coefficient for Customer.service.calls indicates that customers who contacted customer service more frequently were more likely to leave the telecom service, which aligns with the earlier exploratory data analysis findings.

The model also produced two undefined coefficients due to singularities, specifically for International.plan.Yes and Voice.mail.plan.Yes, which occurred because one-hot encoding created perfectly correlated complementary variables. This is expected in logistic regression when both dummy categories are included simultaneously and does not necessarily invalidate the model. Additionally, the reduction from the null deviance (2212.2) to the residual deviance (1729.6) suggests that the predictor variables improved the model's ability to explain customer churn outcomes compared to a model without predictors.

```r
# Predicted probabilities
log_probs <- stats::predict(
  log_model,
  newdata = test_encoded,
  type = "response"
)

# Convert probabilities into class predictions
log_preds <- ifelse(log_probs > 0.5, 1, 0)

# Convert to factor
log_preds <- factor(log_preds,
                    levels = c(0,1))

# Actual values
actual_values <- factor(
  test_encoded$Churn,
  levels = c(0,1)
)

# Confusion matrix
cm_log <- confusionMatrix(log_preds, actual_values, positive = "1")

# Calculate F1-score
f1_score <- 2 * (
  (as.numeric(cm_log$byClass["Pos Pred Value"]) *
   as.numeric(cm_log$byClass["Sensitivity"])) /

  (as.numeric(cm_log$byClass["Pos Pred Value"]) +
   as.numeric(cm_log$byClass["Sensitivity"]))
)

f1_score
```

```
## [1] 0.2442748
```

```r
cm_log
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   0   1
##          0 552  79
##          1  20  16
##
##                Accuracy : 0.8516
##                  95% CI : (0.8223, 0.8777)
##     No Information Rate : 0.8576
##     P-Value [Acc > NIR] : 0.6944
##
##                   Kappa : 0.1801
```

```
##
## Mcnemar's Test P-Value : 5.569e-09
##
##             Sensitivity : 0.16842
##             Specificity : 0.96503
##          Pos Pred Value : 0.44444
##          Neg Pred Value : 0.87480
##              Prevalence : 0.14243
##          Detection Rate : 0.02399
##    Detection Prevalence : 0.05397
##       Balanced Accuracy : 0.56673
##
##        'Positive' Class : 1
##
```

The logistic regression model achieved an overall accuracy of 85.16%; however, this performance was primarily influenced by the majority retained-customer class rather than effective churn detection. The model produced a recall/sensitivity of 16.84%, meaning that only a small proportion of actual churned customers were correctly identified, while the precision value of 44.44% indicates that less than half of the customers predicted to leave actually churned. Additionally, the high specificity of 96.50% shows that the model was highly effective at correctly identifying retained customers, whereas the balanced accuracy of 56.67% suggests uneven predictive performance between retained and churned customer classes due to class imbalance.

Furthermore, the F1-score of 24.43%, which represents the harmonic balance between precision and recall, indicates weak overall churn detection capability despite the model's high general accuracy. These findings suggest that the logistic regression model favored predictions toward retained customers and struggled to effectively identify high-risk churn cases, indicating that additional refinement techniques or alternative machine learning models may be necessary to improve churn prediction performance.

```r
# Confusion matrix table
cm <- table(
  Predicted = log_preds,
  Actual = actual_values
)

# Convert to dataframe
cm_df <- as.data.frame(cm)

# Plot confusion matrix
ggplot(cm_df,
       aes(x = Actual,
           y = Predicted,
           fill = Freq)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Freq),
            size = 6) +
  scale_x_discrete(
    labels = c(
      "0" = "Retained Customer",
      "1" = "Customer Left"
    )
  ) +
  scale_y_discrete(
    labels = c(
      "0" = "Retained Customer",
```

```r
      "1" = "Customer Left"
    )
  ) +
  labs(
    title = "Figure 6. Confusion Matrix for Logistic Regression",
    x = "Actual Class",
    y = "Predicted Class",
    fill = "Count"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )
```

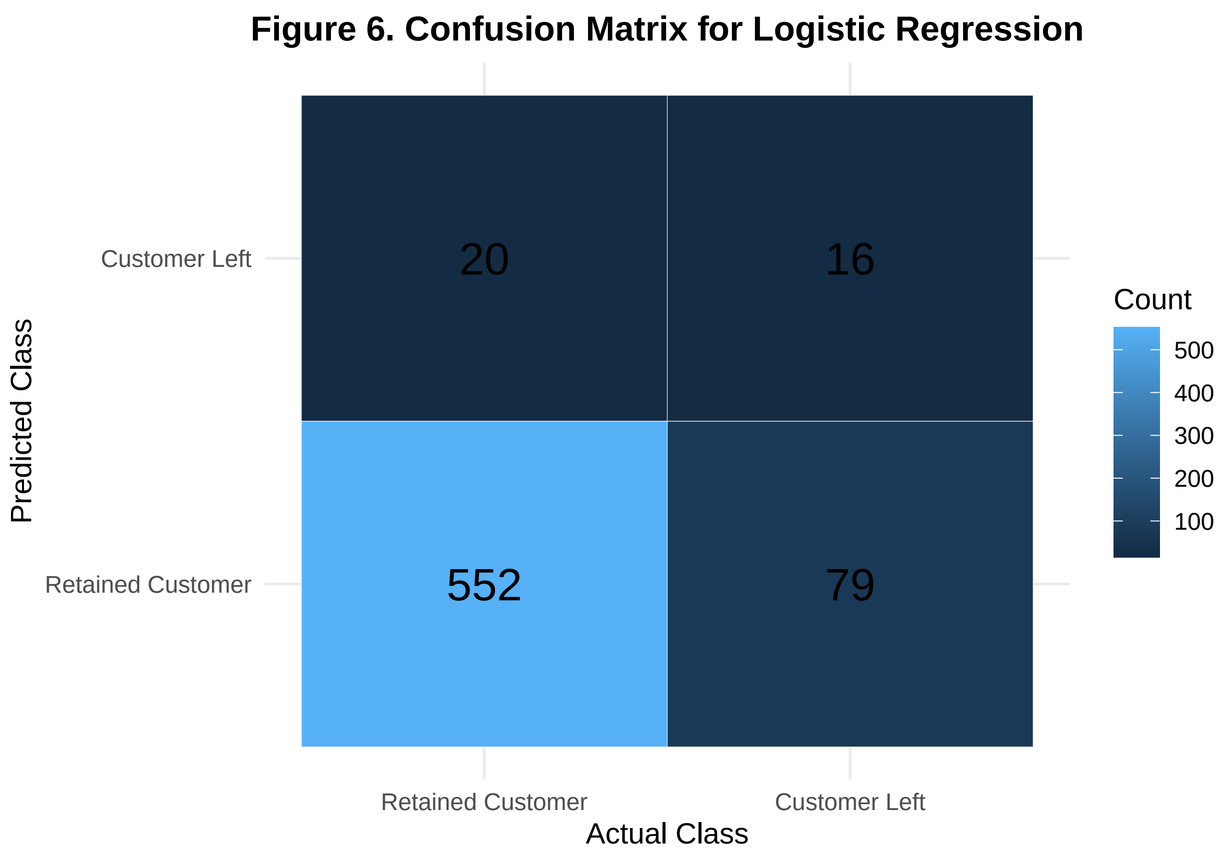
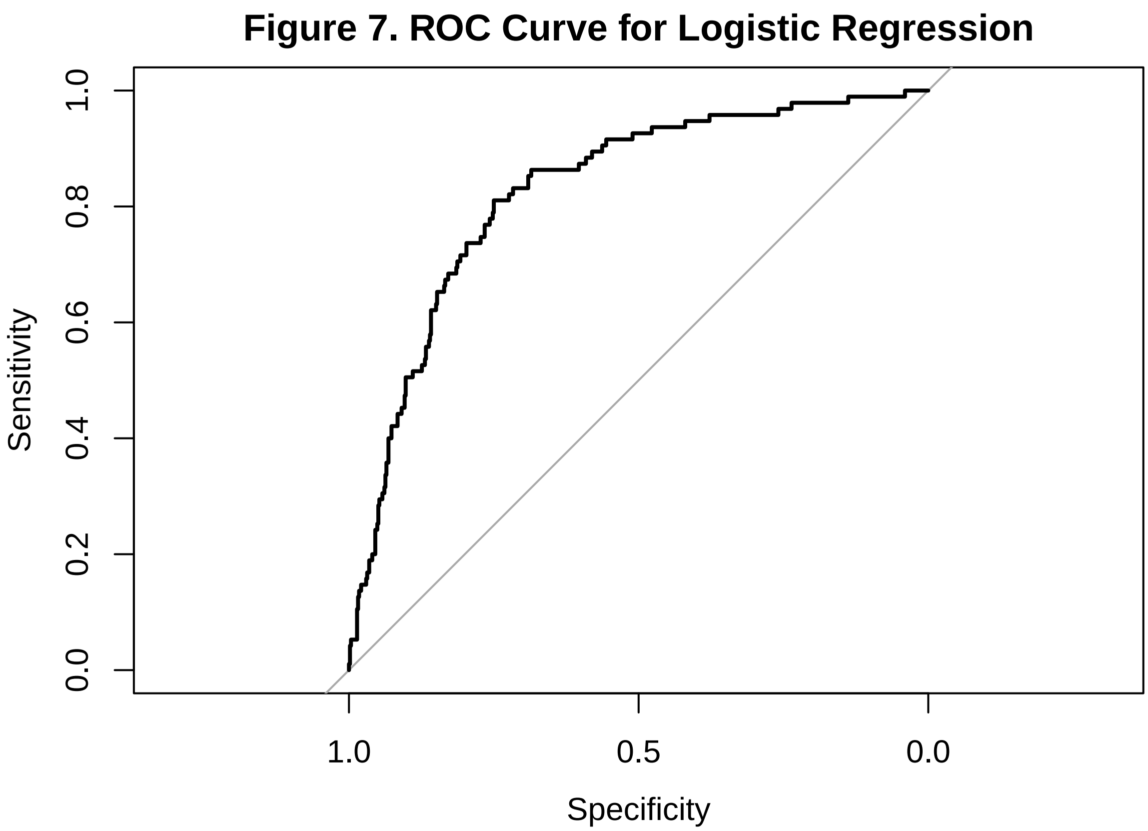


Figure 6 further illustrates that the model was highly effective at predicting retained customers but less effective at detecting customer departures due to the class imbalance present in the dataset. This suggests that the logistic regression model may be biased toward the majority retained-customer class, potentially reducing its ability to accurately detect high-risk churn customers.

```r
# ROC object
roc_obj <- roc(
  actual_values,
  as.numeric(log_probs)
)
```

```r
# Plot ROC curve
plot(
  roc_obj,
  main = "Figure 7. ROC Curve for Logistic Regression"
)
```




```r
# AUC value
auc(roc_obj)
```

```
## Area under the curve: 0.826
```

```r
# AUC value
log_auc <- auc(roc_obj)

log_auc
```

```
## Area under the curve: 0.826
```

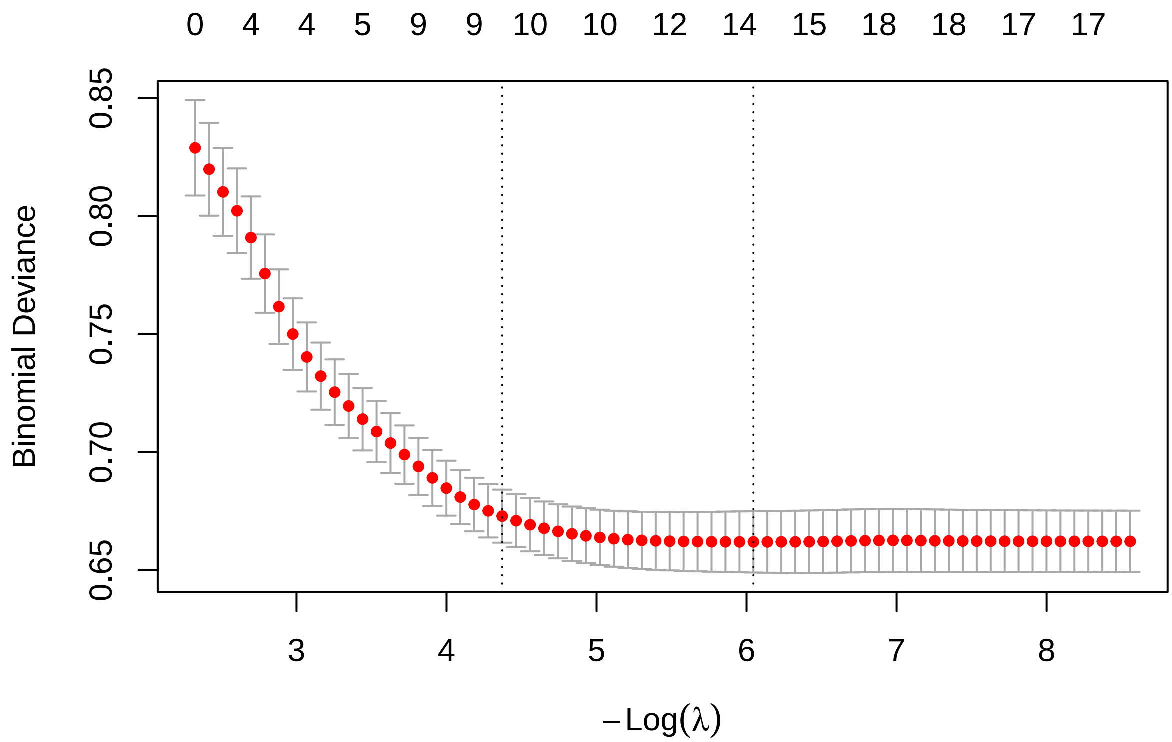
Figure 7 illustrates the Receiver Operating Characteristic (ROC) curve for the logistic regression model, which evaluates the model's ability to distinguish between retained customers and customers who left the telecom service across varying classification thresholds. The ROC curve remains substantially above the diagonal reference line, indicating that the model performs considerably better than random chance in identifying churn outcomes and demonstrates good discriminative capability. Additionally, the model achieved an ROC-AUC value of 0.826, suggesting that the logistic regression model was reasonably effective at ranking customers according to their likelihood of churn.

The steep upward movement of the curve at lower false positive rates suggests that the model can correctly identify a substantial proportion of churn cases before significantly increasing incorrect churn predictions.

Overall, the ROC curve and AUC result indicate that the logistic regression model possesses moderate-to-good classification performance, although improvements may still be necessary to strengthen churn detection effectiveness and reduce the misclassification of customers who leave the service.

Lasso Regression was selected because it improves logistic regression by applying regularization, which helps reduce overfitting and automatically performs feature selection by shrinking less important predictor coefficients toward zero. This makes the model useful for identifying the most influential variables associated with customer churn while improving generalization performance on unseen data.

```r
# Prepare Predictor Matrices

# Predictor matrices
x_train <- as.matrix(
  train_encoded %>%
    dplyr::select(-Churn)
)

x_test <- as.matrix(
  test_encoded %>%
    dplyr::select(-Churn)
)

# Numeric target variable
y_train <- as.numeric(as.character(train_encoded$Churn))
y_test <- as.numeric(as.character(test_encoded$Churn))

# Cross-validation for best lambda
cv_lasso <- cv.glmnet(
  x_train,
  y_train,
  family = "binomial",
  alpha = 1
)

# Plot cross-validation results
plot(cv_lasso)
```




```r
# Best lambda value
cv_lasso$lambda.min
```

```
## [1] 0.002368189
```

```r
lasso_model <- glmnet(
  x_train,
  y_train,
  family = "binomial",
  alpha = 1,
  lambda = cv_lasso$lambda.min
)

# Model coefficients
coef(lasso_model)
```

```
## 20 x 1 sparse Matrix of class "dgCMatrix"
##                                    s0
## (Intercept)            -6.013670e+00
## Account.length          2.717151e-04
## International.plan.No  -2.016607e+00
## International.plan.Yes  1.767080e-04
## Voice.mail.plan.No      8.111835e-01
## Voice.mail.plan.Yes    -7.897673e-06
## Number.vmail.messages   .
## Total.day.minutes       1.181703e-02
## Total.day.calls         1.607426e-03
```

```
## Total.day.charge        5.663822e-04
## Total.eve.minutes       4.917297e-03
## Total.eve.calls         .
## Total.eve.charge        .
## Total.night.minutes     2.163660e-03
## Total.night.calls       4.963883e-04
## Total.night.charge      .
## Total.intl.minutes      8.471777e-02
## Total.intl.calls       -1.031842e-01
## Total.intl.charge       1.761078e-02
## Customer.service.calls  4.802525e-01
```

```r
# Predicted probabilities
lasso_probs <- predict(
  lasso_model,
  s = cv_lasso$lambda.min,
  newx = x_test,
  type = "response"
)

# Convert to class predictions
lasso_preds <- ifelse(lasso_probs > 0.5, 1, 0)

# Convert to factor
lasso_preds <- factor(
  lasso_preds,
  levels = c(0,1)
)

# Actual values
lasso_actual <- factor(
  y_test,
  levels = c(0,1)
)

cm_lasso <- confusionMatrix(
  lasso_preds,
  lasso_actual,
  positive = "1"
)

# Precision
lasso_precision <- as.numeric(
  cm_lasso$byClass["Pos Pred Value"]
)

# Recall
lasso_recall <- as.numeric(
  cm_lasso$byClass["Sensitivity"]
)

# F1-score
lasso_f1 <- 2 * (
  (lasso_precision * lasso_recall) /
```

```r
  (lasso_precision + lasso_recall)
)

lasso_f1
```

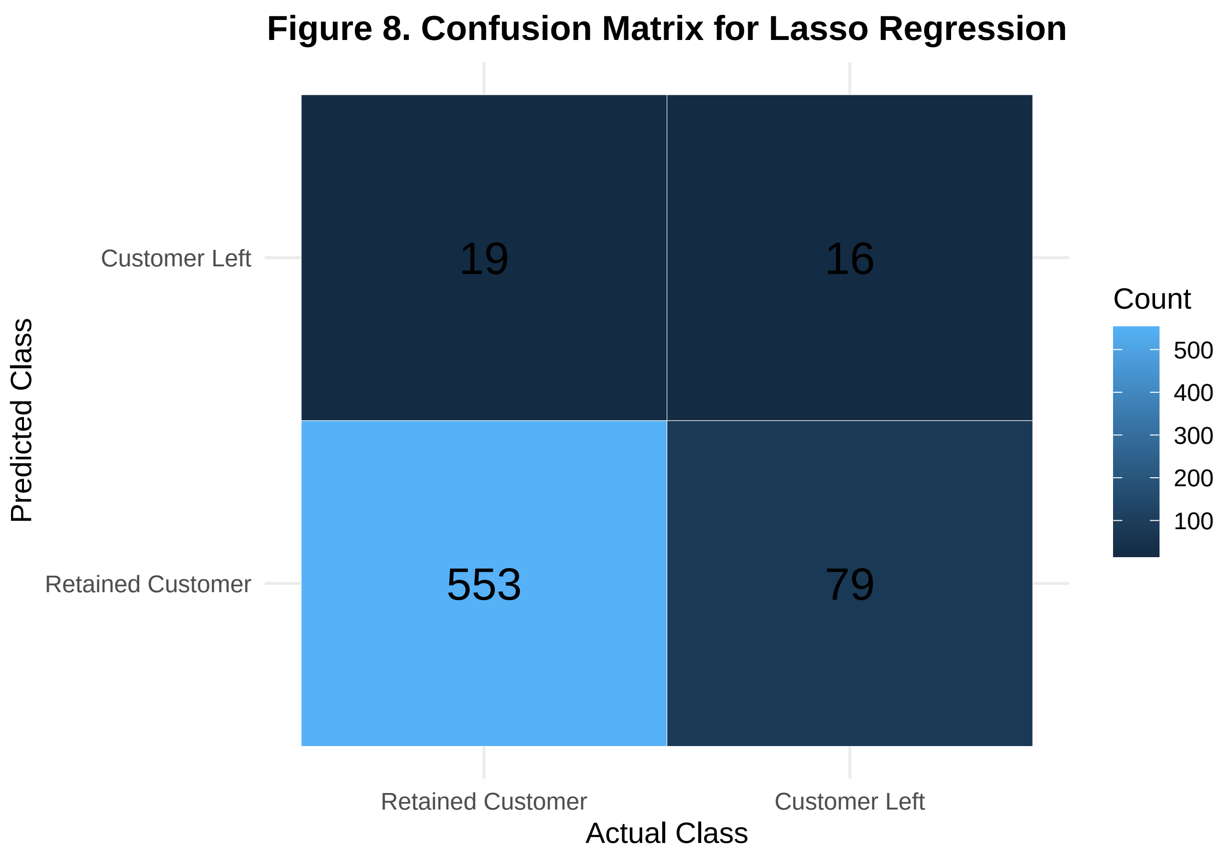
```
## [1] 0.2461538
```

```r
cm_lasso
```

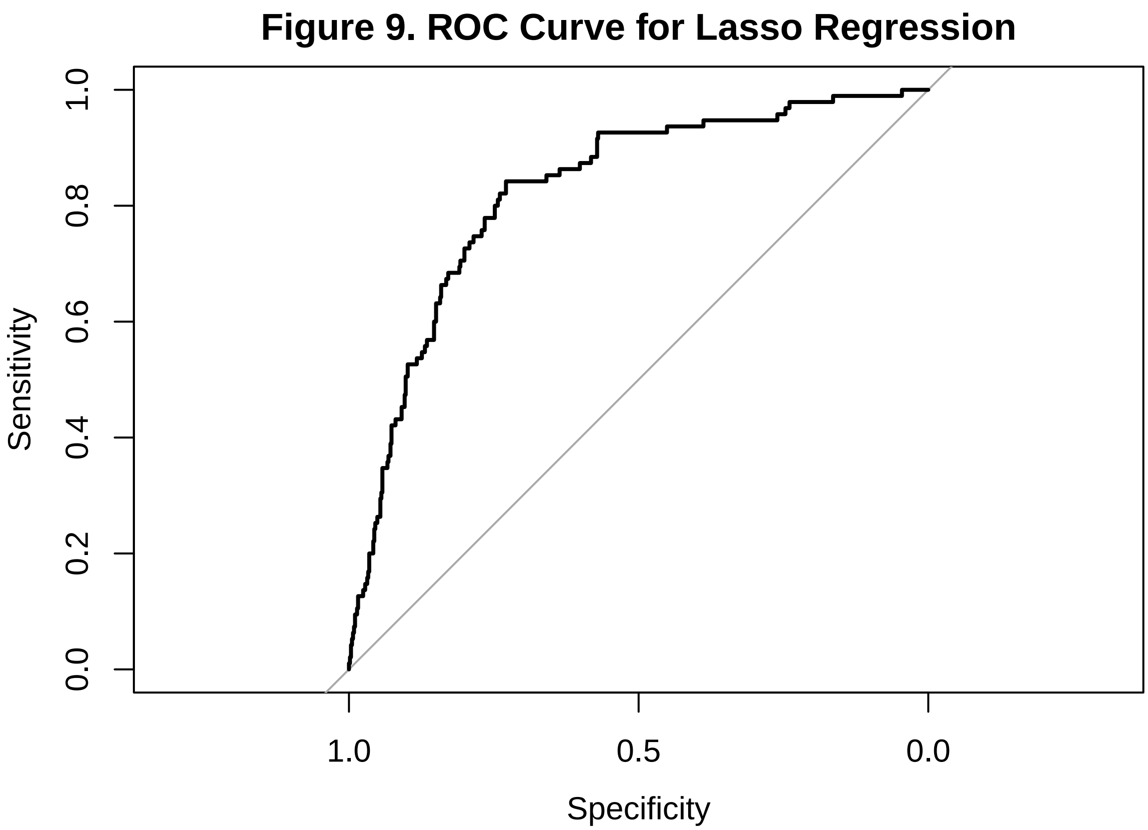
```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   0   1
##          0 553  79
##          1  19  16
##
##                Accuracy : 0.8531
##                  95% CI : (0.8239, 0.8791)
##     No Information Rate : 0.8576
##     P-Value [Acc > NIR] : 0.655
##
##                   Kappa : 0.1835
##
##  Mcnemar's Test P-Value : 2.524e-09
##
##             Sensitivity : 0.16842
##             Specificity : 0.96678
##          Pos Pred Value : 0.45714
##          Neg Pred Value : 0.87500
##              Prevalence : 0.14243
##          Detection Rate : 0.02399
##    Detection Prevalence : 0.05247
##       Balanced Accuracy : 0.56760
##
##        'Positive' Class : 1
##
```

```r
# Confusion matrix table
cm_lasso_table <- table(
  Predicted = lasso_preds,
  Actual = lasso_actual
)

# Convert to dataframe
cm_lasso_df <- as.data.frame(cm_lasso_table)

# Plot confusion matrix
ggplot(cm_lasso_df,
       aes(x = Actual,
           y = Predicted,
           fill = Freq)) +

  geom_tile(color = "white") +

  geom_text(aes(label = Freq),
```

```
          size = 6) +

  scale_x_discrete(
    labels = c(
      "0" = "Retained Customer",
      "1" = "Customer Left"
    )
  ) +

  scale_y_discrete(
    labels = c(
      "0" = "Retained Customer",
      "1" = "Customer Left"
    )
  ) +

  labs(
    title = "Figure 8. Confusion Matrix for Lasso Regression",
    x = "Actual Class",
    y = "Predicted Class",
    fill = "Count"
  ) +

  theme_minimal() +

  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )
```




```r
# ROC curve
lasso_roc <- roc(
  lasso_actual,
  as.numeric(lasso_probs)
)

plot(
  lasso_roc,
  main = "Figure 9. ROC Curve for Lasso Regression"
)
```




```r
# AUC
auc(lasso_roc)
```

```
## Area under the curve: 0.8252
```

3. Model Results / Evaluation

4. Discussion

5. Conclusion

B. Deep Learning Classification Task

Title: Neural Networks and Logistic Regression on the Default Dataset

1. Introduction

2. Methodology

2.1 Data Description

2.2 Data Preprocessing

2.3 Exploratory Data Analysis

2.4 Model Building

2.5 Model Evaluation

3. Results

4. Discussion

5. Conclusion

References:

GitHub repository link: